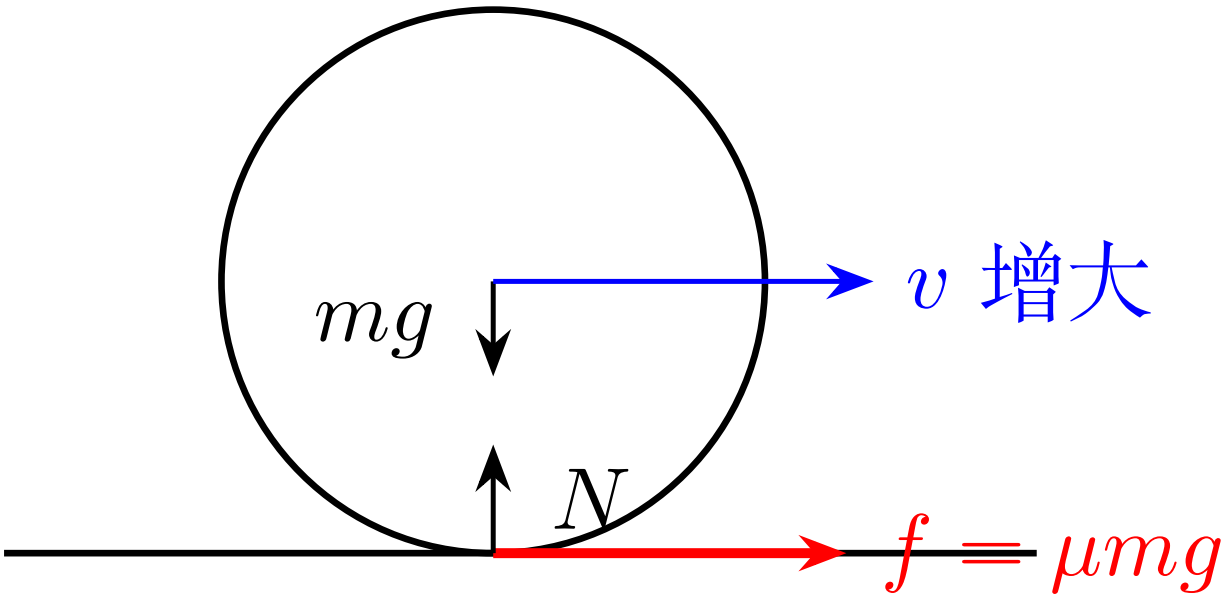ω 减小，$v - R\omega$ 趋零